

IT24101148

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PS_lab7

```
IT24101148_PS_LAB_07.R x
Source on Save Run Source
1 setwd("C:\\Users\\UsEr\\OneDrive\\Desktop\\PS\\IT24101148_PS_Lab_7")
2
3 # Exercise
4 # 1) Uniform Distribution - random variable X follows a uniform distrib
5 punif(25,min = 0, max = 40,lower.tail = TRUE) - punif(10,min = 0, max
6
7
8 # 2) Exponential Distribution
9 pexp(2,rate = 0.33,lower.tail = TRUE)
10
11
12 # 3)
13 # i)
14 # mean = 100 , sd = 15
15 1 - pnorm(130 , mean = 100, sd = 15, lower.tail = TRUE)
16 pnorm(130 , mean = 100, sd = 15, lower.tail = FALSE)
17
18
19 # ii)
20 qnorm(0.95, mean=100, sd=15, lower.tail=TRUE)
21
```

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R 4.5.1 · C:/Users/UsEr/OneDrive/Desktop/PS/IT24101148_PS_Lab_7/
> setwd("C:\\Users\\UsEr\\OneDrive\\Desktop\\PS\\IT24101148_PS_Lab_7")
>
>
> # Exercise
> # 1) Uniform Distribution - random variable X follows a uniform distributi
on with a = 0 and b = 30.
> punif(25,min = 0, max = 40,lower.tail = TRUE) - punif(10,min = 0, max = 4
0, lower.tail = TRUE )
[1] 0.375
>
>
>
> # 2) Exponential Distribution
> pexp(2,rate = 0.33,lower.tail = TRUE)
[1] 0.4831487
>
>
>
> # 3)
> # i)
> # mean = 100 , sd = 15
> 1 - pnorm(130 , mean = 100, sd = 15, lower.tail = TRUE)
[1] 0.02275013
>
>
> # ii)
> qnorm(0.95, mean=100, sd=15, lower.tail=TRUE)
[1] 124.6728
>
```